PROGRAMMING FUNDAMENTALS

LABORATORY ASSIGNMENT — 04

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Section: D

Department: BS Computer Science

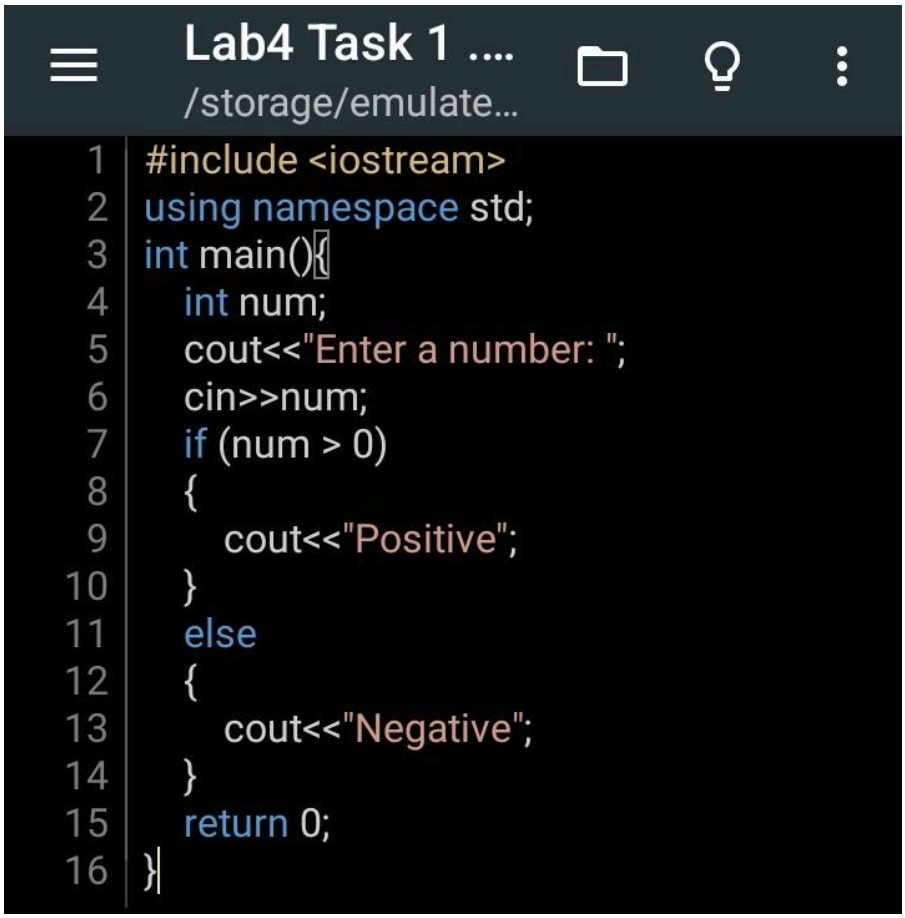
Subject: Programming Fundamentals

Lab: PF Lab 4

Task – 01: (Check Positive or Negative)

Write a program to check the number. Take a number as input from the user. If the number is positive, print "Number is Positive". Otherwise, print "Number is Negative".

Program:



Output:



TASK-01 (Check Number)

Program:

```
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter a number: ";
    cin>>num;
    if (num>0)
    { cout<<"Positive"; }
    else
    { cout<<"Negative"; }
    return 0;
}
```

Output

```
Enter a number: 100
Positive
```

Task – 02: (Number Sign Check)

Program:

Lab4 Task 1 ....
/storage/emulate...

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4     int num;
5     cout<<"Enter a number: ";
6     cin>>num;
7     if (num > 0)
8     {
9        cout<<"Positive";
10    }
11    else if(num<0)
12    {
13       cout<<"Negative";
14    }
15    else
16    {
17       cout<<"Zero";
18    }
19    return 0;
20 }
```

Output:

TAB

```
Enter a number: -12
Negative
[Program finished]
```

TASK - 02 (Sign Check)

Program:

```
#include <iostream>
using namespace std;
int main(){
   int num;
   cout << " Enter a number: "
   cin >> num;
   if (num > 0)
   { cout << " Positive"; }
   else if (num < 0)
   { cout << " Negative"; }
    else
   { cout << "Zero"; }
    return 0;
}
```

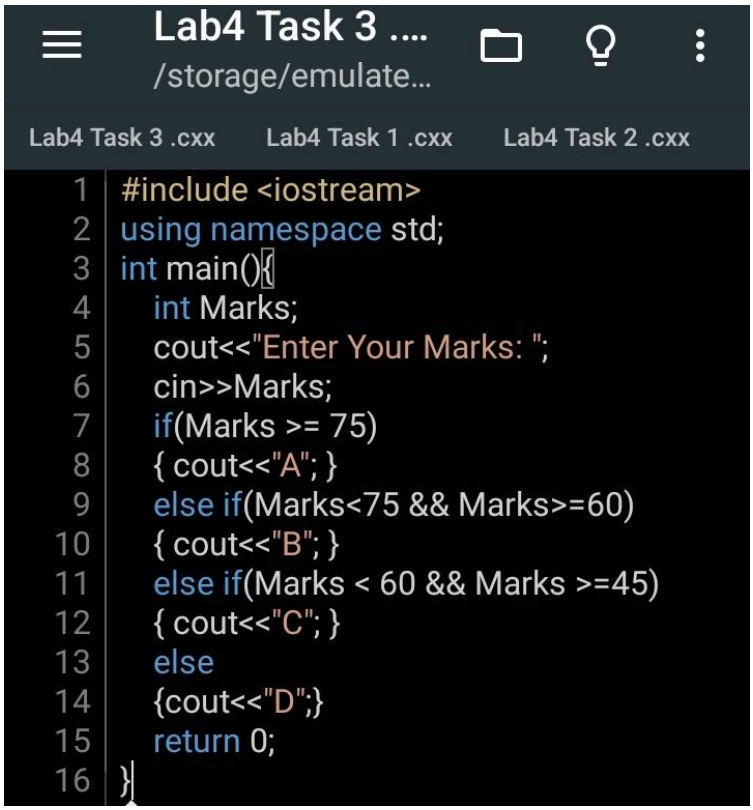
Output

```
Enter a number: -12
Negative
```

Task – 03: (Grading System)

Input : Mark
Process : If mark greater than and equal to 75, score will be A
If mark less than 75 and greater than and equal to 60, score will be B
If mark less than 60 and greater than and equal to 45, score will be C
If mark less than 30, score will be D.

Program:



Output:



TASK - 03 (Grading System)

Program:

```
#include <iostream>
using namespace std;
int main(){
  int Marks;
  cout << "Enter Your Marks: ";
  cin >> Marks;
  if (Marks >= 75)
  {cout << "A";}
  else if (Marks < 75 && Marks >= 60)
  {cout << "B";}
  else if (Marks < 60 && Marks >= 45)
  {cout << "C";}
  else
  {cout << "D";}
  return 0;
}
```

Output

```
Enter your Marks: 45
D
```

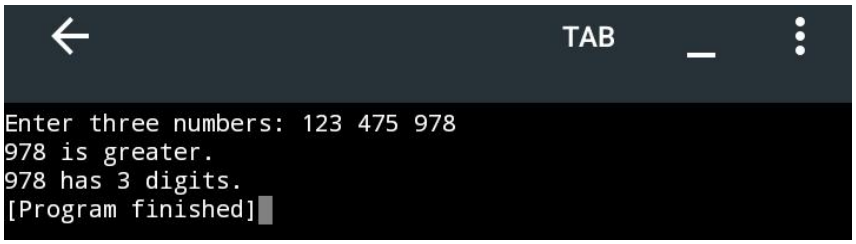
Task – 04: (Largest Number & Digit Count)

Find Largest Number Using Nested if...else statement.
Check whether the number entered by the user is positive or not. If it is positive then calculate how many digits the number has.

Program:

```cpp
#include <iostream>
using namespace std;
int main(){
   int a, b, c, largest;
   cout<<"Enter three numbers: ";
   cin>>a>>b>>c;
   if (a>b){
    if(a>c) {cout<<a<<" is greater.";
largest=a; }
   else { cout<<c<<" is greater."; largest=c;}
   }
   else {
      if(b>c)
   { cout<<b<<" is greater."; largest=b; }
   else
   { cout<<c<<" is greater."; largest=c; }
   }
   if(largest >=0 && largest<=9)
   {cout<<"\n"<<largest<<" has 1 digit.";}
   else if (largest >=10 && largest<=99)
   {cout<<"\n"<<largest<<" has 2 digits.";}
   else if (largest >=100 && largest<=999)
   {cout<<"\n"<<largest<<" has 3 digits.";}
   else if (largest>999)
   {cout<<"\n"<<largest<<" has more than 3
digits.";}
   else
   { cout<<"Not Positive."; }
   return 0;
}
```

Output:



TASK-04 (Largest Number & Digit Count)

Program:

```
#include <iostream>
using namespace std;
int main(){
   int a, b, c, Largest;
   cout << "Enter three numbers: ";
   cin >> a >> b >> c;
   if (a >b)
   { if(a>c){ cout >>a<< " is greater."; Largest=a;}
     else {cout <<c << " is greater."; largest = c;}
    }
    else
    { if (b>c)
     { cout << b << " is greater."; Largest =b;}
    else { cout << c << " is greater."; Largest=c;}
     }
     if(Largest >=0 && largest <=9)
      {cout << "\n" << largest << " has 1 digit.";}
     else if (Largest >=10 && Largest <= 99)
      { cout << "\n" << largest<< " has 2 digits.";}
     else if (Largest > =100 && largest <=999)
      {cout << "\n"<< largest << " has 3 digits.";}
     else if (largest > 999)
      {cout << "\n" << largest << " has more than 3 digits.";}
     else { cout << "Not Positive."}
      return 0;
}
```

Output

```
Enter three numbers: 123 475 978
978 is greater.
978 has 3 digits.
```